Geometric and Analytic Aspects of Elliptic Functions, Modular Forms, and Picard's Theorem

Ali ADALI

27 November 2006

Abstract

This paper explores elliptic functions and modular forms from a geometric perspective, beginning with Riemann's mapping theorem and conformal classifications of Riemann surfaces. We thoroughly develop the analytic properties of elliptic functions using the Weierstrass $\wp$-function, construct the λ and J modular functions, and examine the structure of the modular group. Finally, these foundational tools are applied to prove Picard's celebrated theorem on the values attained by non-constant entire functions.

1. Introduction

In this paper we begin with observing the appearance of elliptic functions and modular forms from a geometric point of view. Our motivation begins with Riemann's mapping theorem, which may be stated as any simply connected Riemann surface is conformally equivalent to the Riemann sphere, complex plane, or upper half plane, and this theorem can be extended to every Riemann surface (compact or not) in a proper way. This result suggests that the determination of conformal automorphisms is important. The automorphisms of the Riemann sphere give almost trivial results, whereas those of the complex plane case or the upper half plane case are more interesting, which lead connections to elliptic functions and modular forms respectively. After these observations, we will study elliptic functions thoroughly. We'll prove some basic analytic properties and construct the elliptic functions with the help of Weierstrass's $\wp$ function. These results will be followed by constructing λ and J modular functions in order to give an introduction for defining modular forms and to use certain properties that will be used in the last section. The modular group and modular forms will be defined extensively, and in the last section, the celebrated theorem of Picard will be proved in order to show the relevance of modular forms to analytic functions with a nice application.

Riemann Mapping Theorem and Conformal Classifications of Riemann Surfaces

Riemann's mapping theorem states that any simply connected Riemann surface R can be mapped conformally (and, hence, homeomorphically) onto a domain D of exactly one of the following types:

i) $D = \mathbb{C} \cup \{\infty\}$, the extended complex plane;

ii) $D = \mathbb{C}$, the finite complex plane; or

iii) $D = \{z \in \mathbb{C} : |z| < 1\}$, the unit disc.

Corollary 1 (Riemann) *Every Riemann surface (compact or not) is a factor of one of the domains i, ii, iii by a discrete subgroup acting freely.*

In the case of an arbitrary, not necessarily simply connected Riemann surface, its type is the same as the type of its universal covering surface, which is always simply

connected. Therefore we may consider the universal covering $\tilde{X}$ instead of the Riemann surface X.

From the topological point of view, these surfaces i, ii, iii correspond to Riemann surfaces of genus 0, genus 1, and genus ≥ 1 respectively.

The conformal automorphisms of these domains are:

i) For $D = \mathbb{C} \cup \{\infty\} = S^2$: the Möbius group $z \mapsto \frac{az+b}{cz+d}$; $a, b, c, d \in \mathbb{C}$.

ii) Affine group $z \mapsto az + b$ with $a \neq 0$ (fixing ∞).

iii) $PSL_2(\mathbb{R}) : z \mapsto \frac{az+b}{cz+d}$, $a, b, c, d \in \mathbb{R}$ (fixing the real line).

Elliptic Curves and Elliptic Functions

Definition 1 *A non-singular complete algebraic curve of genus 1.*

Elliptic curves are isomorphic to Riemann surfaces, which can be identified by $\mathbb{C}^2/L$ for some lattice L in $\mathbb{C}^2$. The correspondence between elliptic curves and elliptic functions comes from the fact that the functions are defined on L. The geometric correspondence, however, is as follows:

Let X be an elliptic curve over an algebraically closed field k. Then X is biregularly isomorphic to a plane cubic curve. If $\mathrm{char}(k) \neq 2, 3$, then in the projective plane there is an affine coordinate system in which the equation of X is in normal Weierstrass form:

$$y^2 = x^3 + ax + b \tag{1}$$

The curve X is non-singular if and only if the polynomial $x^3 + ax + b$ does not have multiple zeros, that is, if the discriminant $\Delta = -16(4a^3 + 27b^2) \neq 0$. The j-invariant of an elliptic curve X,

$$j(X) = 1728\frac{4a^3}{4a^3 + 27b^2}, \tag{2}$$

does not depend on the choice of the coordinate system. Two elliptic curves have the same j-invariant if and only if they are biregularly isomorphic.

We shall study elliptic functions extensively.

Definition 2 (Elliptic functions) *Elliptic functions are doubly periodic meromorphic functions, where doubly periodic functions are functions having two periods ω_1, ω_2 such that ω_1/ω_2 is non-real.*

We have some properties and facts about these functions. First of all, we call (ω_1, ω_2) a fundamental pair if every period of an elliptic function f is of the form $m\omega_1 + n\omega_2$ for some integers m, n. The first fact is that for every f there exists a fundamental pair of periods. Let us define the lattice $\Omega = \{m\omega_1 + n\omega_2 \mid n, m \in \mathbb{Z}\}$. If pairs (ω_1, ω_2) and (ω_1', ω_2') generate the same lattice Ω, then $\omega_2' = a\omega_2 + b\omega_1'$ and $\omega_1' = c\omega_2 + d\omega_1'$ where $ad - bc = \pm 1$. Some other properties are:

Lemma 1 *If an elliptic function f has no poles or zeros in some period parallelogram, then f is constant.*

Proof. If it has no poles in the period parallelogram, then f is continuous, hence bounded on the closure of the parallelogram. By periodicity, f is bounded everywhere, and by Liouville's theorem, f is constant. Checking $1/f$ gives the result when f has no zeros on the period parallelogram.

Lemma 2 *The contour integral of an elliptic function along the boundary of any cell (a translated parallelogram with no poles or zeros on its boundary) is zero.*

Proof. The integrals along parallel edges cancel because of periodicity.

Lemma 3 *The sum of the residues of an elliptic function at its poles in any period parallelogram is zero.*

Proof. Applying Cauchy's theorem to a cell and using Lemma 2.

Lemma 4 *The number of zeros of an elliptic function in any period parallelogram is equal to the number of poles, each counted with multiplicity.*

Proof. The integral $\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}dz$, taken around the boundary C of a cell, counts the difference between the number of zeros and the number of poles inside the cell. But f'/f is elliptic with the same periods as f, and by Lemma 2 this integral is zero.

Construction of Elliptic Functions

We want to find the simplest elliptic functions with given periods. Since the order of such a function is at least 2, we need a second-order pole or 2 simple poles in each period parallelogram. We shall follow the classical example of Weierstrass, who chose a function with a double pole.

For simplicity, we place the pole at the origin, and since multiplication by a constant is irrelevant, we may require that the singular part is z^{-2}. As f is elliptic, it must be an even function, because $f(z) - f(-z)$ has the same period and no singularity, thus it must reduce to a constant, which is zero by setting $z = \omega_1/2$.

Now our aim is to prove that

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \neq 0} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right) \tag{3}$$

is the desired elliptic function, where the sum ranges over all $\omega = m\omega_1 + n\omega_2$ except 0. Our first task is to check that the series converges. If $|\omega| > 2|z|$, then

$$\left| \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right| = \left| \frac{z(2\omega - z)}{\omega^2 (z-\omega)^2} \right| \leq \frac{10|z|}{|\omega|^3} \tag{4}$$

Therefore the series converges, uniformly on every compact set, as

$$\sum_{\omega\neq 0}\frac{1}{|\omega|^3} \tag{5}$$

converges. This convergence can be verified as follows: since ω_1/ω_2 is non-real, there exists a $k>0$ such that $|n_1\omega_1+n_2\omega_2|\geq k(|n_1|+|n_2|)$ for all real pairs (n_1,n_2). Considering integers, there are $4n$ pairs with $|n_1|+|n_2|=n$. This gives

$$\sum_{\omega\neq 0}|\omega|^{-3}\leq 4k^{-3}\sum_{1}^{\infty}n^{-2}<\infty. \tag{6}$$

The next step is to prove that the right-hand side of the equation has the periods ω_1,ω_2, but this is obvious, because $\wp'(z)=-2\sum_\omega\frac{1}{(z-\omega)^3}$. We have shown that this function has periods ω_1,ω_2, thus $\wp'(z)=\wp'(z+\omega)$ for each period. Therefore $\wp(z+\omega)-\wp(z)$ is constant, which is (as calculated above) zero, hence $\wp$ has the required periods.

We have further properties of $\wp$:

Theorem 1 *$\wp$ satisfies the differential equation*

$$[\wp'(z)]^2=4\wp(z)^3-60G_4\wp(z)-140G_6, \tag{7}$$

where $G_n=\sum_{\omega\neq 0}\omega^{-n}$ for $n\geq 3$.

Proof. Near $z=0$ we have:

$$\wp'(z)=-\frac{1}{z^3}+6G_4z+20G_6z^3+\ldots, \tag{8}$$

an elliptic function of order 3. Its square has order 6 since:

$$[\wp'(z)]^2=\frac{4}{z^6}-\frac{24G_4}{z^2}-80G_6+\ldots \tag{9}$$

where $+\ldots$ indicates power series in z vanishing at 0. Now

$$4\wp^3(z)=\frac{4}{z^6}+\frac{36G_4}{z^2}+60G_6+\ldots \tag{10}$$

hence

$$[\wp'(z)]^2-4\wp^3(z)=-\frac{60G_4}{z^2}-140G_6+\ldots \tag{11}$$

so

$$[\wp'(z)]^2-4\wp^3(z)+60G_4\wp(z)=-140G_6+\ldots \tag{12}$$

Since the left member has no pole at $z=0$, it has no poles anywhere in a period parallelogram, so it must be constant. Therefore, the constant must be $-140G_6$, which proves the theorem.

In addition to these, we have the property of $\wp$ that every even elliptic function f is of the form

$$f(z) = C\frac{\prod_{k=1}^{n}[\wp(z) - \wp(a_k)]}{\prod_{r=1}^{m}[\wp(z) - \wp(b_r)]} \tag{13}$$

and for an odd elliptic function f, $f(z)/\wp'(z)$ is even and has the same periods as f, so it is of the form above. Therefore we may write every elliptic function f in the form

$$f(z) = R_1[\wp(z)] + \wp'(z)R_2[\wp(z)] \tag{14}$$

where R_1, R_2 are rational functions of $\wp$ having the same periods as f.

The Modular Function $\lambda(\tau)$

The differential equation in Theorem 1 can also be written as

$$(\wp'(z))^2 = 4(\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3) \tag{15}$$

where e_1, e_2, e_3 are roots of $4\omega^2 - g_2\omega - g_3$.

By symmetry and periodicity, $\wp(\omega_1 - z) = \wp(z)$, hence $\wp'(\omega_1 - z) = -\wp'(z)$, from which it follows that $\wp(\omega_1/2) = 0$. Similarly, $\wp(\omega_2/2) = 0 = \wp((\omega_2 + \omega_1)/2)$. The numbers $\omega_1, \omega_2, (\omega_1 + \omega_2)/2$ are incongruent modulo periods, thus they are three zeros of $\wp'$, which is of order 3, and all zeros are simple. Comparing to the above, we may set that

$$e_1 = \omega_1, \quad e_2 = \omega_2, \quad e_3 = (\omega_1 + \omega_2)/2. \tag{16}$$

These roots are all distinct because $\wp$ assumes each e_k with multiplicity 2, and if 2 of them were equal, then that value would be assumed 4 times, in contradiction with $\wp$ being of order 2. By this definition, we have that e_k are homogeneous of order -2 in ω_1, ω_2. We conclude that the quantity

$$\lambda(\tau) = \frac{e_3 - e_2}{e_1 - e_2} \tag{17}$$

depends only on the ratio $\tau = \omega_2/\omega_1$. It is clear from the definition of $\wp$ that $\lambda(\omega)$ is the quotient of two analytic functions in the upper half plane. Because $e_1 \neq e_2$ and $e_3 \neq e_2$, it is never 0 or 1.

If we change the periods ω_1, ω_2 by a unimodular transformation, then $\wp$ doesn't change, so e_1, e_2, e_3 can at most permute. Setting ω_1', ω_2' as new periods: the case $\omega_1'/2 = \omega_1/2, \omega_2'/2 = \omega_2/2$ is trivial. The case $\omega_2'/2 = (\omega_1 + \omega_2)/2, \omega_1'/2 = \omega_1/2$ means e_2 and e_3 are interchanged, hence λ goes over into $\lambda/(\lambda - 1)$. In the last case, $\omega_2'/2 = \omega_1/2, \omega_1'/2 = \omega_2/2$, so e_2 and e_1 are interchanged, and λ goes over into $1 - \lambda$. Sample transformations are $\tau \to \tau + 1$ and $\tau \to -1/\tau$. We find that $\lambda(\tau)$ satisfies the functional equations:

$$\lambda(\tau + 1) = \frac{\lambda(\tau)}{\lambda(\tau) - 1}, \quad \lambda\left(-\frac{1}{\tau}\right) = 1 - \lambda(\tau) \tag{18}$$

The Conformal Mapping by $\lambda(\tau)$

For convenience we shall henceforth use the normalization $\omega_1 = 1, \omega_2 = \tau$. With this choice of periods, we obtain from the previous relations:

$$e_3 - e_2 = \sum_{m,n=-\infty}^{\infty} \left[\frac{1}{(m - 1/2 + (n + 1/2)\tau)^2} - \frac{1}{(m + (n - 1/2)\tau)^2} \right] \tag{19}$$

$$e_1 - e_2 = \sum_{m,n=-\infty}^{\infty} \left[\frac{1}{(m - 1/2 + n\tau)^2} - \frac{1}{(m + (n - 1/2)\tau)^2} \right] \tag{20}$$

where the double series are absolutely convergent. Our first observation is that these quantities are real when τ is purely imaginary (this is also true of the individual e_k). Indeed, when we replace τ by $-\tau$, the sums remain the same, except for a rearrangement of the terms. We conclude that $\lambda(\tau)$ is real on the imaginary axis.

Because $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ is the congruence subgroup mod 2, we have $\lambda(\tau + 2) = \lambda(\tau)$. In other words, $\lambda(\tau)$ has period 2, which means that $\lambda(\tau)$ can be expressed as a function of $e^{\pi i \tau}$.

To evaluate the sum above, using

$$\frac{\pi^2}{\sin^2(\pi z)} = \sum_{n=-\infty}^{\infty} \frac{1}{(z - n)^2}, \tag{21}$$

we obtain:

$$e_3 - e_2 = \pi^2 \sum_{n=-\infty}^{\infty} \left(\frac{1}{\cos^2 \pi(n - 1/2)\tau} - \frac{1}{\sin^2 \pi(n - 1/2)\tau} \right) \tag{22}$$

and

$$e_1 - e_2 = \pi^2 \sum_{n=-\infty}^{\infty} \left(\frac{1}{\cos^2 \pi n\tau} - \frac{1}{\sin^2 \pi n\tau} \right) \tag{23}$$

The series are strongly convergent, both for $n \to \infty$ and $n \to -\infty$, for $|\cos(n\pi\tau)|$ and $|\sin(n\pi\tau)|$ are comparable to $e^{|n|\pi \mathrm{Im}(\tau)}$; the convergence is uniform for $\mathrm{Im}\,\tau > 0$.

The limits can now be taken termwise, and we find that $e_3 - e_2 \to 0$ and $e_1 - e_2 \to \pi^2$ (from the term $n = 0$). Hence $\lambda(\tau) \to 0$ as $\mathrm{Im}\,\tau \to \infty$, uniformly with respect to the real part of τ. It follows further by the second equation that $\lambda(\tau) \to 1$ when τ approaches 0 along the imaginary axis.

We need one more thing, namely the order to which $\lambda(\tau)$ vanishes together with $e^{\pi i \tau}$. From the equations above, the leading terms in $e_3 - e_2$ are the ones corresponding to $n = 0$ and $n = 1$. The sum of these terms is

$$2\pi^2 \left[\frac{4e^{\pi i\tau}}{(1 + e^{\pi i\tau})^2} + \frac{4e^{\pi i\tau}}{(1 - e^{\pi i\tau})^2} \right] \tag{24}$$

and we conclude that

$$\lambda(\tau)e^{-\pi i\tau} \to 16 \quad \text{for } \operatorname{Im}\tau \to \infty. \tag{25}$$

The region Ω is defined such that it is bounded by the imaginary axis, the line $\operatorname{Re}\tau = 1$, and the circle $|\tau - 1/2| = 1/2$. The transformation $\tau + 1$ maps the imaginary axis onto $\operatorname{Re}\tau = 1$, and $1 - 1/\tau$ maps $\operatorname{Re}\tau = 1$ onto $|\tau - 1/2| = 1/2$. Since $\lambda(\tau)$ is real on the imaginary axis, it follows by virtue of the functional relations that it is real on the boundary of Ω. Furthermore, $\lambda(\tau) \to 1$ as τ tends to 0 and $\lambda(\tau) \to \infty$ as τ tends to 1 inside Ω.

We apply the argument principle to determine the number of times $\lambda(\tau)$ takes a non-real value ω_0 in Ω. Cut off the corners of Ω by means of a horizontal line segment $\operatorname{Im}\tau = t_0$ and its images under $-1/\tau$ and $1 - 1/\tau$. For sufficiently large t_0, it is clear that $\lambda(\tau) \neq \omega_0$ in the portions that have been cut off. The circle near $\tau = 1$ is mapped by $\lambda(\tau)$ on a curve $\lambda(\tau) = \lambda(1 - 1/\tau) = 1 - 1/\lambda(\tau)$, where $\tau = s + it_0, 0 \leq s \leq 1$; in view of the asymptotic behavior, this is approximately a large semicircle in the upper half plane. It is now clear that the image of the contour of the truncated region Ω has winding number 1 about ω_0 if $\operatorname{Im}\omega_0 > 0$ and winding number 0 otherwise. As a result, $\lambda(\tau)$ takes every value in the upper half plane exactly once in Ω, and no value in the lower half plane. This is also sufficient to guarantee that $\lambda(\tau)$ is monotone on the boundary of Ω, because if it were not, $\lambda'(\tau)$ would vanish at a boundary point, and it would be impossible for a full semicircular neighborhood of that boundary point to be mapped into the upper half plane. As a result, we conclude the following theorem:

Theorem 2 *The modular function $\lambda(\tau)$ effects a one-to-one conformal mapping of the region Ω onto the upper half plane. The mapping extends continuously to the boundary in such a way that $\tau = 0, 1, \infty$ corresponds to $\lambda = 1, \infty, 0$.*

The Klein's Modular J Function

Definition 3 *Since ω_1/ω_2 is not real, we define*

$$J(\omega_1, \omega_2) = \frac{g_2^3(\omega_1, \omega_2)}{\Delta(\omega_1, \omega_2)} \tag{26}$$

where $\Delta = g_2^3 - 27g_3^2$ (since the discriminant is never zero, this is well-defined).

Since g_2^3 and Δ are homogeneous of the same degree (degree 12), we have $J(\lambda\omega_1, \lambda\omega_2) = J(\omega_1, \omega_2)$; in particular, if $\tau \in \mathbb{H}$, we have $J(1, \tau) = J(\omega_1, \omega_2)$. Thus $J(\omega_1, \omega_2)$ is a function of the ratio τ alone. We write $J(\tau)$ for $J(1, \tau)$.

We have invariance properties of J under modular transformations due to the following statements.

If ω_1, ω_2 are given periods with non-real ratio, we introduce new periods ω_1', ω_2' by the relations $\omega_2' = a\omega_2 + b\omega_1$ and $\omega_1' = c\omega_2 + d\omega_1$ where a, b, c, d are integers with $ad - bc = 1$.

Then the pair (ω_1', ω_2') is equivalent to (ω_1, ω_2) in that it generates the same set of periods Ω. Therefore $g_2(\omega_1, \omega_2) = g_2(\omega_1', \omega_2')$ and $g_3(\omega_1, \omega_2) = g_3(\omega_1', \omega_2')$ since g_2, g_3 depend only on Ω. Consequently, $J(\omega_1, \omega_2) = J(\omega_1', \omega_2')$.

The ratio of the new periods is

$$\tau' = \frac{\omega_2'}{\omega_1'} = \frac{a\omega_2 + b\omega_1}{c\omega_2 + d\omega_1} = \frac{a\tau + b}{c\tau + d}. \tag{27}$$

An easy calculation shows that

$$\mathrm{Im}(\tau') = \mathrm{Im}\left(\frac{a\tau + b}{c\tau + d}\right) = \frac{\mathrm{Im}(\tau)}{|c\tau + d|^2}. \tag{28}$$

Hence $\tau' \in \mathbb{H}$ if and only if $\tau \in \mathbb{H}$. The equation $\tau' = \frac{a\tau+b}{c\tau+d}$ is called a *unimodular transformation* if a, b, c, d are integers with $ad - bc = 1$. The set of all unimodular transformations forms a group (under composition) called the modular group. This group and such functions will be discussed further below. The foregoing remarks show that J is invariant under the transformations of the modular group; that is, we have:

Theorem 3 *If $\tau \in \mathbb{H}$ and a, b, c, d are integers with $ad - bc = 1$, then $\frac{a\tau+b}{c\tau+d} \in \mathbb{H}$ and*

$$J\left(\frac{a\tau + b}{c\tau + d}\right) = J(\tau). \tag{29}$$

Modular Group and Modular Forms

To give a geometrical background: The only conformal automorphisms of the Riemann sphere onto itself are the linear fractional transformations:

$$T : z \mapsto \frac{az + b}{cz + d} \tag{30}$$

where a, b, c, d are complex numbers and $ad - bc = 1$. The set of all linear fractional transformations forms a group which is isomorphic to the group of complex 2×2 matrices of determinant 1 (denoted by $SL(2, \mathbb{Z})$) divided by its center $\pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Each linear transformation except the identity has at most two fixed points, and at least one fixed point. The group of all linear fractional transformations acts transitively on the sphere; indeed, there is a unique linear fractional transformation which sends any given ordered triple of distinct points into any other given such triple. A linear fractional transformation also transforms circles on the Riemann sphere into circles.

We shall consider only linear fractional transformations with real coefficients; these map the upper half plane onto itself. Any such transformation is one of the following three types:

1. **Elliptic transformation:** This transformation has two fixed points $\zeta, \bar{\zeta}$ with ζ in the upper half plane. After a change of variable sending $(\zeta, \bar{\zeta})$ into $(0, \infty)$, the transformation takes the form $\omega' = K\omega$, where $K = e^{i\theta}$.

2. **Hyperbolic transformation:** This transformation has two fixed points on the real line. These are sent into 0 and ∞ by a suitable change of variable, and the transformation takes the form $\omega' = K\omega$, where $K > 0$.

3. **Parabolic transformation:** This transformation has only one fixed point, either ∞ or on the real line. By sending the fixed point to ∞ if necessary, the transformation takes the form $\omega' = \omega + c$.

We shall be interested in analytic functions on $\mathbb{H}$ which have invariance properties under some group of linear fractional transformations.

Definition 4 *The modular group, Γ, is the group of linear fractional transformations $L : z \mapsto \frac{az+b}{cz+d}$ with $ad - bc = 1$ and a, b, c, d integers. (There are in fact two groups to be considered here: first is the group $\Gamma' = SL(2, \mathbb{Z})$ of 2×2 matrices with integer coefficients and determinant 1, called the homogeneous modular group. The group of distinct linear fractional transformations is the quotient group $\Gamma \simeq \Gamma'/(\pm I)$, which is the inhomogeneous modular group.)*

By the assertions above, we see that elliptic transformations of Γ occur when $a + d = 0$ or $a + d = \pm 1$. Therefore, a simple calculation shows that the first leads to a transformation L for which $L^2 = I$, and the second case leads to $L^3 = I$.

Definition 5 *$z_1, z_2 \in \mathbb{H}$ are called equivalent under a group G of transformations (denoted by $z_1 \sim_G z_2$) if there is a $T \in G$ with $z_1 = Tz_2$.*

This is clearly an equivalence relation. A fundamental domain for G is an open set D which does not contain any pairs of distinct equivalent points and whose point-set closure contains at least one point from each equivalence class. Our task is now to determine a fundamental domain for Γ. We have the following theorem:

Theorem 4 *A fundamental domain for Γ is the set*

$$D = \{z \in \mathbb{H} \mid |Re\, z| < 1/2, |z| > 1\}. \tag{31}$$

Proof. We first show that D is the same set as

$$D_1 = \{z \in \mathbb{H} \mid |\mathrm{Re}\, z| < 1/2, |cz + d| > 1\} \tag{32}$$

for all pairs of integers except $c = d = 0$. Setting $c = 1, d = 0$ shows that $D \subset D_1$. Conversely, suppose that $z \in D$; then

$$|cz + d|^2 = (cx + d)^2 + c^2y^2 = c^2(x^2 + y^2) + 2cdx + d^2 > c^2 - cd + d^2 \geq 1 \tag{33}$$

for all (c,d) except $(0,0)$. Now from earlier remarks it follows that the closure of D contains at least one point from each equivalence class under Γ. To show that the pairs which are equivalent on the closure of D are on the boundary: these points are identified by $T : z \mapsto z+1$ and $S : z \mapsto -1/z$. Suppose $z, z' \in D$ satisfy $z \sim_\Gamma z'$, say $z' = Lz$; then $\operatorname{Im} z = \operatorname{Im} Lz$, so that

$$1 = |cz+d|^2 = (cx+d)^2 + c^2y^2 \geq c^2 - cd + d^2 \geq 1 \tag{34}$$

hence either $c = 0, d = \pm 1$ or $d = 0, c = \pm 1$. Calculating these, we have that the first case is $L = T$ and the second is $L = S$; these are identifications already mentioned, thus concluding the proof.

Notation 1 *We will use the following notations: $\mathbb{H}$ denotes the upper half plane; G denotes a subgroup of finite index μ in Γ, where Γ is the inhomogeneous modular group; and $\mathbb{H}/G$ denotes the identification space of $\mathbb{H}$ modulo G, considered as a Riemann surface.*

It is of interest to study those meromorphic functions on $\mathbb{H}$ which are invariant under all transformations in G, i.e., $f(Tz) = f(z)$ for all $T \in G$. However, this is too restrictive for many purposes. Instead we shall consider those meromorphic functions such that $f(z)$ and $f(Tz)$ have the same zeros and poles. In that case, $\frac{f(Tz)}{f(z)} = \mu_T(z)$ is a holomorphic function on $\mathbb{H}$ that is never zero. There is moreover a consistency condition satisfied by the class of functions $\{\mu_T(z) \mid T \in G\}$: $\mu_{TS}(z) = \mu_T(Sz)\mu_S(z)$ for $T, S \in G$. It follows from the chain rule and the fact that the maps are analytic automorphisms that these two requirements are satisfied by the classes:

$$\mu_T(z) = J_T(z)^k = \left(\frac{dT}{dz}\right)^k. \tag{35}$$

It can be shown [1] that these classes of functions are, essentially, the only ones satisfying these two conditions. Recalling that $J_T(z) = \frac{dT}{dz} = (cz+d)^{-2}$ if $T : z \mapsto \frac{az+b}{cz+d}$, we make the following definition:

Definition 6 *An unrestricted modular form of weight k for G is a meromorphic function $f(z)$ on $\mathbb{H}$ such that*

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^{2k} f(z) = J_T(z)^{-k} f(z)$$

for all $T \in G$, where k is an integer.

In order to give an extended definition for modular forms, we shall make the following definitions as well:

Definition 7 *The modular form $f(z)$ is holomorphic at ∞ if $\hat{f}(\zeta)$ is holomorphic in $|\zeta| < 1$, where $\zeta = e^{2\pi i z/q}$ and $\hat{f}(\zeta) = f(z)$. In particular, $\hat{f}(\zeta)$ has a Taylor expansion in ζ:*

$$\hat{f}(\zeta) = \sum_{m=0}^{\infty} a_m \zeta^m \tag{36}$$

This induces a Fourier expansion for f:

$$f(z) = \sum_{m=0}^{\infty} a_m e^{2\pi m z/q} \tag{37}$$

We call a_0 the value of f at ∞; if a_n is the first non-zero coefficient, n is called the order of f at ∞.

Definition 8 *Let p be a parabolic fixed point of G, not ∞. Let $S \in G$ map p into ∞, and let g be the S-transform of f. Then f is holomorphic at p if g is holomorphic at ∞. The value and order of f at p are those of g at ∞.*

Remark: The above definition is independent of S. In fact, if S' is another transformation satisfying our hypothesis and g' is the S'-transform of f, then $g'(z) = g(z + c)$ for some integer c. From the fact that g is holomorphic at ∞, the desired result follows.

Now we may define a modular form:

Definition 9 *A modular form is an unrestricted modular form which is holomorphic at all points of $\mathbb{H}$ and all parabolic vertices of the group.*

An Application: Proof of Picard's Theorem

Theorem 5 (Picard's Theorem) *Every non-constant entire function attains every complex value with at most one exception.*

Proof. Let $f(z)$ be an entire function such that f does not take values a and b (where $a \neq b$ are distinct finite values). Consider

$$f_1(z) = (f(z) - a)/(b - a);$$

f_1 is also entire and never takes 0 and 1. Thus, without loss of generality, we may assume $a = 0, b = 1$. By Theorem 6, there exists τ_0 such that $\lambda(\tau_0) = f(0)$. Because $\lambda'(\tau_0) \neq 0$ by the same theorem, there exists a local inverse of λ defined in a neighborhood Δ_0 of $f(0)$, denoted by λ_0^{-1}, satisfying $\lambda_0^{-1}(\lambda(\omega)) = \omega$ in Δ_0 and

$$\lambda_0^{-1}(f(0)) = \tau_0. \tag{38}$$

By continuity, there is a neighborhood of the origin in which $f(z) \in \Delta_0$, and we can therefore define $h(z) = \lambda_0^{-1}(f(z))$ in that neighborhood. We shall let h be the global analytic function obtained by continuing the function element (h, Δ_0) in all possible ways. h is arbitrarily continuable in the entire finite z-plane. By the monodromy theorem, the continuation of h exists as a single-valued function analytic in the entire finite z-plane. Thus h is an entire function, and so too is $\phi(z) = e^{ih(z)}$. But $\operatorname{Im} h(z) > 0$ since $h \in \mathbb{H}$, so $|\phi(z)| = e^{-\operatorname{Im} h(z)} < 1$ (or bounded accordingly). Therefore ϕ is a bounded entire function, and by Liouville's theorem, this must be constant. This implies that h is constant, and hence f is constant, completing the proof.